

SIPOC DIAGRAM

Order Processing & Fulfillment Process | High-Level Process Map

SUPPLIERS	INPUTS	PROCESS	OUTPUTS	CUSTOMERS
Primary Suppliers: - E-commerce order system - Inventory management system - Warehouse management system (WMS) - Shipping/carrier system - Internal staff (pickers, packers, QC) - Inventory and packaging suppliers External Suppliers: - Shipping carriers (UPS, FedEx, USPS) - Material suppliers (boxes, tape, labels)	Data Inputs: - Customer order details - Order specification file - Inventory location data - Shipping address/requirements Physical Inputs: - Products from inventory - Packaging materials - Shipping labels Other Inputs: - Quality standards - Shipping deadlines	Step 1: Order Receipt - Receive customer order in system - Validate order completeness Step 2: Order Verification - Check inventory availability - Verify customer details - Flag any issues Step 3: Picking - Retrieve items from warehouse - Stage items at packing area Step 4: Packing - Pack items in box - Prepare shipping label - Add packing slip Step 5: Quality Check - Verify order accuracy - Check packaging condition - Scan shipment Step 6: Shipping - Hand off to carrier - Update system status - Generate tracking number	Physical Outputs: - Packed & shipped order - Shipping confirmation - Packing materials waste Data Outputs: - Tracking number - Shipping confirmation - Order status update - Performance metrics - Inventory adjustment Quality Indicators: - Order accuracy rate - Shipping timeliness - System status accuracy	Primary Customers: - End customers - Retail partners Internal Customers: - Customer service team - Finance/accounting - Shipping department - Management/leadership Carrier/External: - Shipping carriers - Distribution partners

SIPOC ANALYSIS - KEY INSIGHTS

Critical Suppliers & Dependencies:

- Inventory Management System:** Accuracy of inventory data directly impacts picking accuracy and order fulfillment. Current accuracy: 94% (target: 99.8%)
- Order System:** Data quality at receipt affects downstream processes. Validation at Step 2 currently catching 8% error rate
- Warehouse Staff:** Picking and packing quality depends on staff training and process adherence
- Shipping Carriers:** External dependency; impact on delivery time limited but included for full visibility

Process Bottlenecks Identified:

- Picking Step (Step 3):** Longest cycle time; warehouse layout not optimized for distance
- Verification Step (Step 2):** Manual checking creates delays when issues found
- Quality Check (Step 5):** Currently only spot-checking; missing errors that progress to customers

Key Inputs to Monitor:

- Inventory data accuracy (94% baseline)
- Order specification completeness at receipt
- Shipping deadline clarity

Success Metrics (CTSSs) Aligned to SIPOC:

- **CTS-1 (Cycle Time ≤ 2 days):** Dependent on optimization of Steps 1-6, especially picking
- **CTS-2 (Accuracy ≥ 99.5%):** Dependent on verification (Step 2), picking accuracy (Step 3), quality check (Step 5)
- **CTS-5 (Inventory Accuracy ≥ 99.8%):** Dependent on supplier (inventory system) and process execution

PROCESS MEASUREMENT POINTS

Process Step	Key Metrics to Track	Current Baseline
Step 1: Order Receipt	Orders received/hour; Data completeness %	500 orders/day; 98% complete
Step 2: Verification	Verification time; % orders flagged; Resolution time	0.8 min/order; 3% flagged; 2 hours resolve
Step 3: Picking	Picking time; Picking accuracy; Lines picked/hour	2.1 min/order; 97%; 28 lines/hour
Step 4: Packing	Packing time; Packing accuracy; Orders/hour	0.6 min/order; 98%; 100/hour
Step 5: QC	QC time; Defects found; First-pass yield	0.3 min/order; 2%; 92% FPY
Step 6: Shipping	Handoff time; Carrier pickup time; Tracking accuracy	0.2 min/order; 4 hours avg; 85% accurate
TOTAL CYCLE TIME	End-to-end from receipt to shipment	4.2 days (Target: 2.0 days)

PROCESS SCOPE DEFINITION

Start Point (Process Boundary):

Customer order entry into e-commerce system (order receipt trigger)

End Point (Process Boundary):

Order handed off to shipping carrier and tracking number provided to customer

Excluded from This SIPOC:

- Returns processing (separate process)
- Customer service interactions (parallel process)
- Carrier delivery (external service)
- Payment processing (upstream process)

Note: This SIPOC represents the high-level order fulfillment process. In the Measure Phase, this will be decomposed into lower-level SIPOCs for each critical subprocess (e.g., picking process, packing process) to identify detailed improvement opportunities.

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Next Review: Measure Phase - decompose into subprocess SIPOCs